

# Filters

Based on components

Active filters → Adv: it is possible to gain increase the gain.  
 Passive filters → Reactive N/W, since entire performance depends on reactive elements.

Based on operating freq:-

We have 5 filters based on freq:-

1. Lowpass filter
2. High pass filter
3. Band pass filter
4. Band elimination filter
5. All pass filter.

\* Based on the components present in filters the filters are classified as Active filter & passive filter.

\* "Active filter" consists of operational amplifier & capacitor. Generally inductor is not used in the active filter since size is bulky & cost is high.

\* In the Active filter it is possible to increase gain of the system.

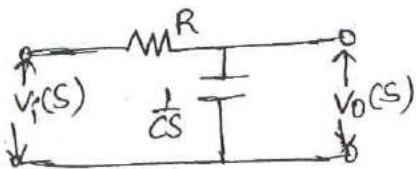
\* passive filter consists of LC series-parallel sections. (reactive N/W)

\* Based on the operating freq. Filters are classified as

1. LPF
2. HPF
3. BPF
4. BRF (or) BSF (or) Band elimination filter
5. APF

Note:  $X_L = 2\pi fL$   
 $X_C = \frac{1}{2\pi fC}$

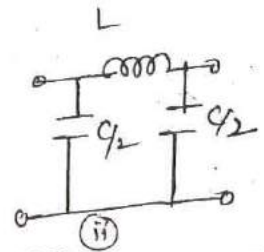
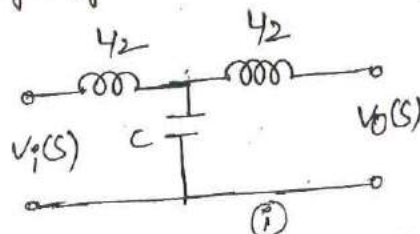
Lowpass filter:- While designing LPF always 'C' is shunt element.



↳ first order LPF

$$\frac{V_o(s)}{V_i(s)} = \frac{X_C}{R + X_C}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{1}{RCs + 1}$$

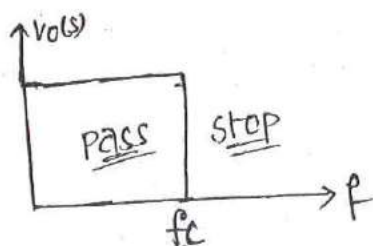


second order LPF  
 ↓  
 Having two energy stored elements

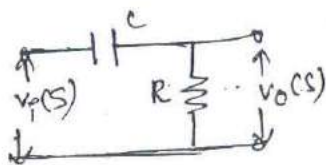
while designing Higher order filters always symmetrical N/Ws are designed.

At  $\omega = 0 \Rightarrow V_o(s) = V_i(s)$   
 At  $\omega = \infty \Rightarrow V_o(s) = 0$  } so it is a lowpass filter

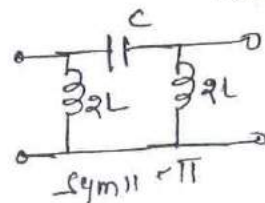
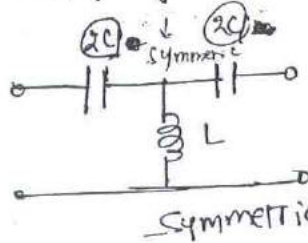
LPF



## 2) High pass filters :-



while designing HPF always capacitor is series element



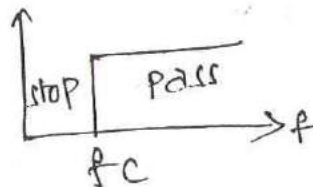
If any low frequency component is present it capacitor offers high impedance, & offers low impedance path for high frequency.

$$V_o(s) = V_i(s) \frac{R}{R + 1/sC}$$

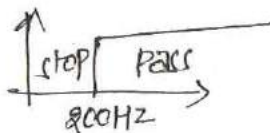
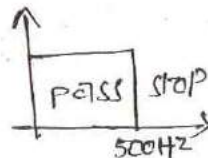
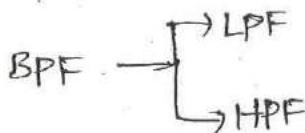
$$\Rightarrow \frac{V_o(s)}{V_i(s)} = \frac{RCs}{RCs + 1}$$

(HPF)

$$\begin{aligned} \text{At } \omega = 0 &\Rightarrow V_o(s) = 0 \\ \omega = \infty &\Rightarrow V_o(s) = V_i(s) \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{so HPF}$$

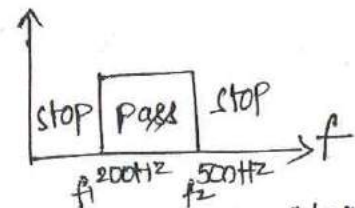
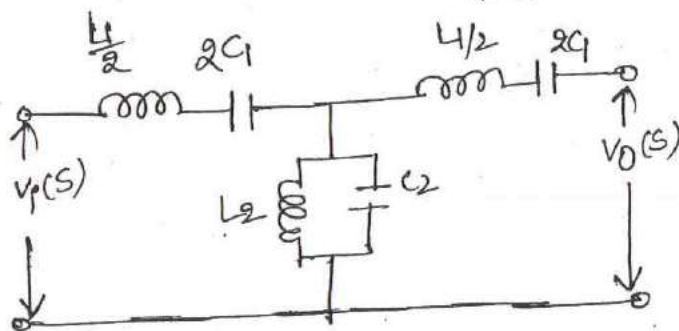


## Bandpass filters :-



condition is :

$$f_{c,LPF} > f_{c,HPF}$$



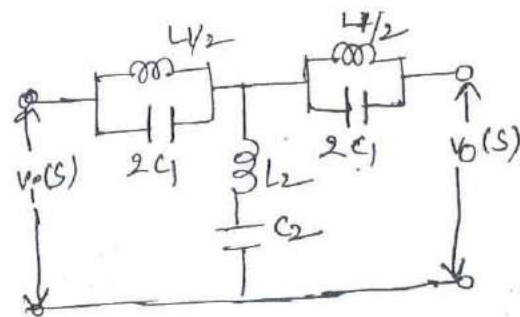
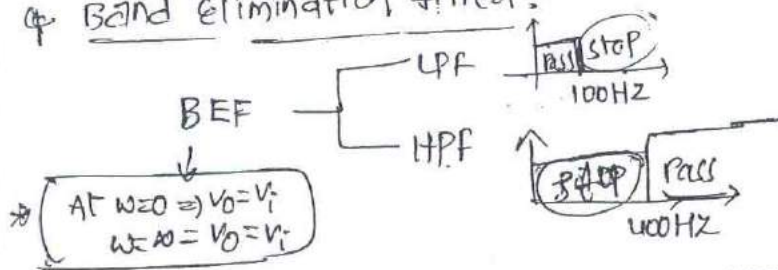
\* BPF is obtained with a combination of LPF & HPF. And cutoff frequency of LPF should be greater than cutoff frequency of HPF.

Band pass filter provides low impedance only for some band of frequency & offers high impedance for all other frequencies.

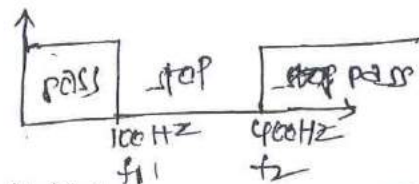
\* In the BPF first cutoff frequency is calculated w.r.t to "series resonance" and second cutoff frequency is calculated w.r.t to "parallel resonance".

$$\begin{aligned} \text{At } \omega = 0 &\Rightarrow V_o = 0 \\ \omega = \infty &\Rightarrow V_o = 0 \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{BPF.}$$

## BAND Elimination filter:-

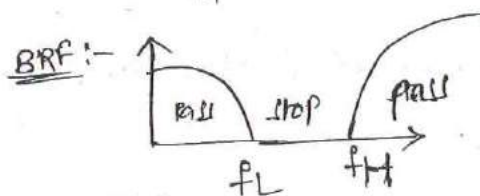
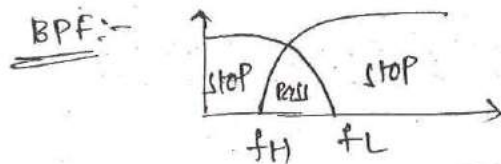
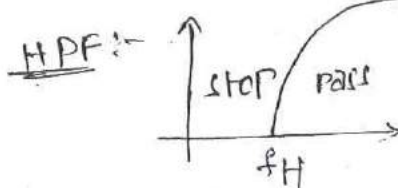


BEF offers High Impedance for some Band of frequ which are reg in b/w  $f_1$  &  $f_2$  as shown in the figure.



\* BEF is obtained with the combination of low pass and high pass filter and cutoff frequ of HPF should be greater than cutoff frequ of LPF.

$$f_{sHPF} > f_{cLPF}$$

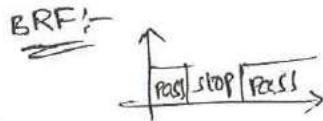


\* \* \*  
BRF = Low pass + High pass  
BPF = High pass + Low pass.

note:-  
BPF, BRF  $\Rightarrow$  Having two cutoff frequency.



ie first stop & then pass  
so HPF first  
pass & stop so LPF second.



ie first passband & then stopband  $\Rightarrow$  LPF  
next pass stopband & then passband  $\Rightarrow$  HPF  
so first LPF next HPF

## First order filter T.F:-

$$\frac{1}{1+TS} \rightarrow \text{LPF} \quad [T=RC]$$

$$\frac{TS}{1+TS} \rightarrow \text{HPF}$$

$$\frac{1-TS}{1+TS} \rightarrow \text{All pass}$$

## second order filter T.F:-

$$\frac{P}{s^2+as+b} \rightarrow \text{LPF}$$

$$\frac{PS^2}{s^2+as+b} \rightarrow \text{HPF}$$

$$\frac{PS}{s^2+as+b} \rightarrow \text{BPF}$$

$$\frac{PS^2+Q}{s^2+as+b} \rightarrow \text{BEF}$$

$$\frac{s^2-PS+Q}{s^2+as+b} \rightarrow \text{All pass}$$

→ which filter having two zeros at the origin

Ans:- HPF not BPF.

like that they are asking the questions.

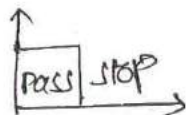
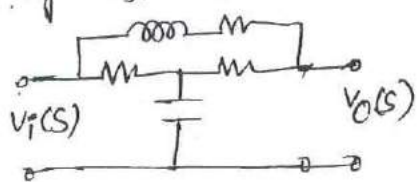
Note:-

for all pass filter

1. zeros are present in the right half of plane & poles are present in the left half of plane.

2. poles & zeros are symmetrical about jw axis.

(pb) identify type of the filter of the N/w shown.



so it is LPF...

Sol:-  $\omega=0 \Rightarrow V_o(s) = V_i(s)$   $\rightarrow$  max means pass band  
 $\omega=\infty \Rightarrow V_o(s) = 0$   $\rightarrow$  stop band.

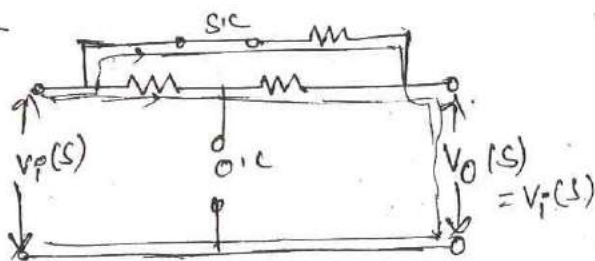
Note:- At  $\omega=0 \Rightarrow \therefore \omega L = 0$  so Inductor short-ckt.

At  $\omega=\infty \Rightarrow$  Inductor open-ckt.

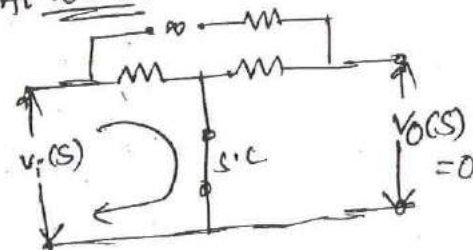
Similarly At  $\omega=0 \Rightarrow X_c = \frac{1}{\omega C} = \infty$  Capacitor is open-circuit

At  $\omega=\infty \Rightarrow$  Capacitor short-ckt.

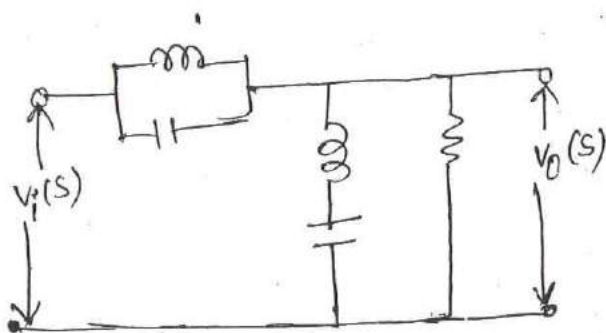
At  $\omega=0$ :-



At  $\omega=\infty$ :-



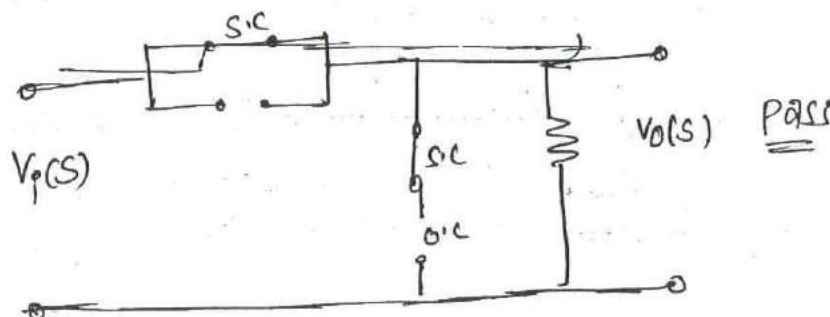
(pb)



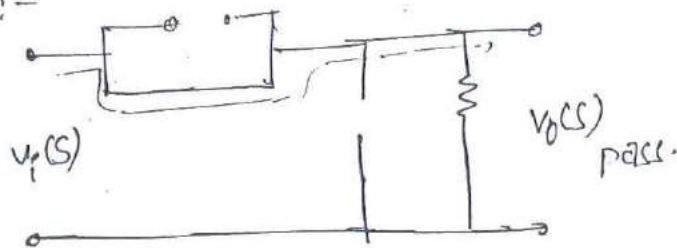
Identify type of filter?

Sol:

At  $\omega=0$ :-



$\omega = \infty$  :-

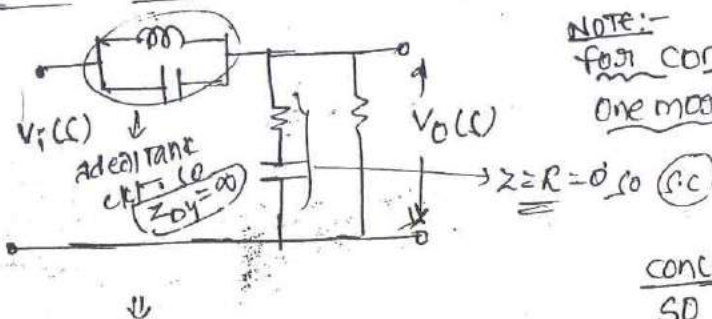


So for  $\omega = 0 \Rightarrow V_o = V_i$

$\omega = \infty \Rightarrow V_o = V_i$

It is a BRF (or) BEF

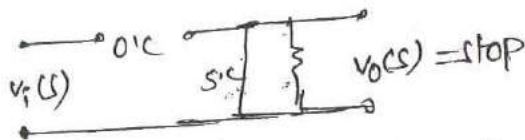
Now check whether it is a BEF (or) All pass filter :-



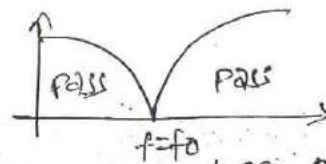
Note :- For conclusion we have to take one more freq i.e. resonant freq ( $f_o$ )

Conclusion :-

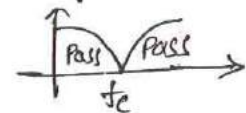
So the given filter is a BEF.



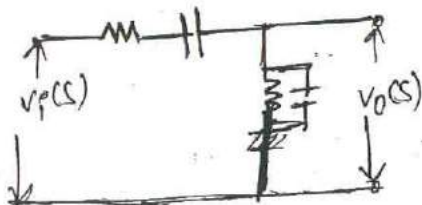
corresponding to this filter  $\rightarrow$



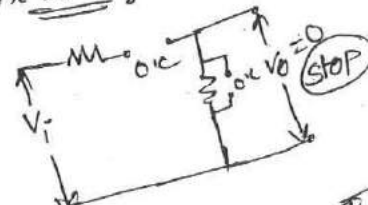
Note :- When Band elimination filter eliminates only few frequencies then it is also called as "Notch filter".



Pb



$\rightarrow$  At  $\omega = 0$  :-

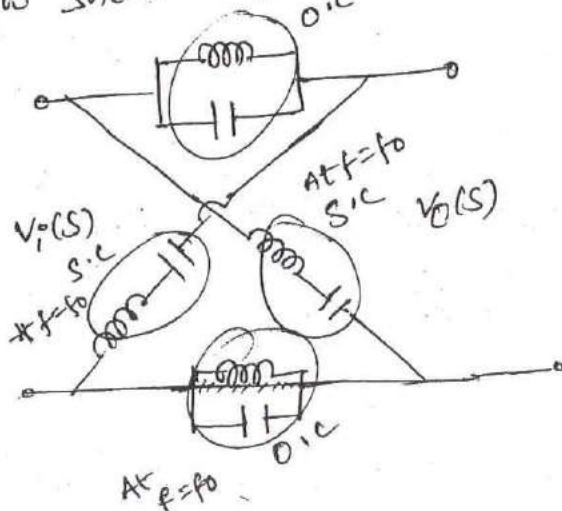


Soln :-

$\omega = 0 \Rightarrow V_o = 0$   
 $\omega = \infty \Rightarrow V_o = 0$  } So it is a BPF.

Pb

Identify type of the filter of the N/w shown.



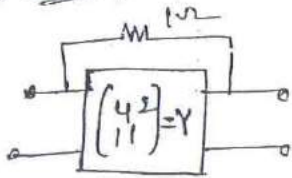
Soln :- At  $\omega = 0$  :-  $V_o = V_i$

$\omega = \infty$  :-  $V_o = V_i$

At  $f = f_o$  :-  $V_o = V_i$

So it is a All pass filter.

Q:1



$[Y] = ?$

Sol:-

$$\begin{aligned} Y &= Y_a + Y_b \\ &= \begin{bmatrix} 4 & 2 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 5 & 1 \\ 0 & 2 \end{bmatrix} \end{aligned}$$

Q:2

$$A = D = \frac{3}{2} B = \frac{4}{3} C$$

$Z_{11} = ?$

$$Z_{11} = \frac{V_1}{I_1} \Big|_{I_2=0}$$

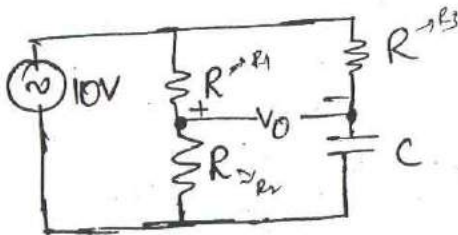
$$\begin{aligned} V_1 &= AV_2 - BI_2 \\ I_1 &= CV_2 - DI_2 \end{aligned}$$

$$\begin{aligned} \frac{V_1}{V_2} &= A \\ \frac{I_1}{V_2} &= C \end{aligned} \Big|_{I_2=0}$$

$$\therefore Z_{11} = \frac{V_1}{I_1} = \frac{V_1/V_2}{I_1/V_2} = \frac{A}{C} = \frac{4/3}{2} = \frac{4}{3}$$

$$\boxed{Z_{11} = \frac{4}{3}}$$

Q:4



$V_0(j\omega) = ?$

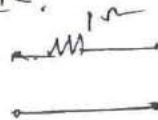
Sol:-

$$V_{R1} = 5V; V_{R2} = 5V$$

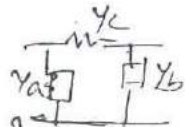
$$V_{R3} = \frac{(10)(R)}{R - j\omega C} //$$

$$V_C = \frac{(10)(-j\omega C)}{R - j\omega C} //$$

note:

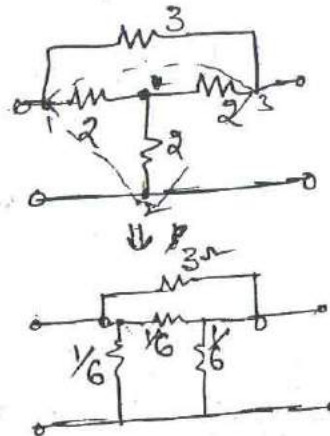


$$\begin{aligned} Y &= \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \\ &= \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \end{aligned}$$



$$\begin{aligned} Y &= \begin{bmatrix} Y_a + Y_c & -Y_c \\ -Y_c & Y_b \end{bmatrix} \\ &= \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \end{aligned}$$

Q:8



$$\begin{aligned} Y &= Y_1 + Y_2 \\ &= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} + \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{bmatrix} \\ &= \begin{bmatrix} \frac{5}{6} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} \end{bmatrix} // \dots \end{aligned}$$

$$\begin{aligned} Y_{21} &= \frac{1}{6} - \frac{1}{3} = -\frac{1}{6} \\ &= -\frac{1}{6} \Omega^{-1} \end{aligned}$$

Q:5  
Sol:-

$$Z_{in} = \frac{V_1}{I_1}$$

$$V_1 = AV_2 + BI_2 = 1V_2 + 2I_2$$

$$I_1 = CV_2 + DI_2 = 1V_2 + 3I_2$$

$$\text{but } V_2 = 10I_2$$

$$\begin{aligned} V_1 &= 12I_2 \Rightarrow \frac{V_1}{I_1} = \frac{12}{13} \Omega \end{aligned}$$

$$\begin{aligned}
 V_0 &= 5 - V_c \\
 &= 5 + \frac{10jX_c}{R - jX_c} \\
 &= \frac{5R + j5X_c}{R - jX_c}
 \end{aligned}$$

so Ans: (B)

$$\begin{aligned}
 |V_0(j\omega)| &= 5 \left| \frac{R + jX_c}{R - jX_c} \right| \\
 &= 5(1)
 \end{aligned}$$

Note:-

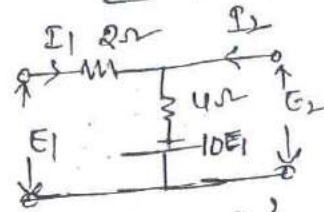
If  $V_0$  is not given then Ans: (D)

Note:-

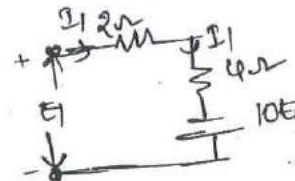
$$\begin{aligned}
 Z_{in} &= \frac{V_1}{I_1} = \frac{AV_2 - BI_2}{CV_2 - DI_2} \\
 &= \frac{A(-I_2 R_L) - BI_2}{-}
 \end{aligned}$$

$$\underline{Q(6)} \quad Z = \begin{bmatrix} 18 & 7 \\ 7 & 28 \end{bmatrix}$$

Q: (7)

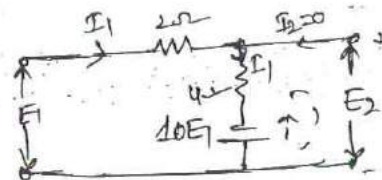


$$\begin{aligned}
 \text{Sol: } Z_{11} &= ?, Z_{21} = ? \\
 Z_{11} &= \frac{V_1}{I_1} \bigg|_{I_2=0}
 \end{aligned}$$



$$\begin{aligned}
 \text{KVL: } -E_1 + 2I_1 + 4I_1 - 10E_1 &= 0 \\
 \Rightarrow \frac{E_1}{I_1} &= \frac{6}{11}
 \end{aligned}$$

$$Z_{21} = \frac{V_2}{I_2} \bigg|_{I_1=0}$$



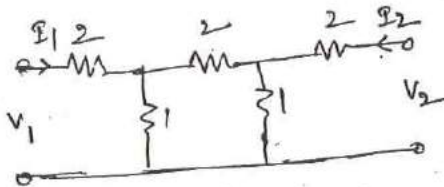
$$V_1 = \frac{6}{11} I_1$$

$$V_{21} = \frac{6}{11} I_1$$

$$\therefore Z_{21} = \frac{-16}{11}$$

$$\begin{aligned}
 10E_1 - 4I_1 + E_2 &= 0 \\
 E_2 &= 4I_1 - 10E_1 \\
 &= 4I_1 - 10\left(\frac{6}{11}I_1\right) \\
 &= \left(\frac{44 - 60}{11}\right)I_1 \\
 E_2 &= \frac{-16}{11}I_1 \\
 \Rightarrow Z_{21} &= \frac{-16}{11}
 \end{aligned}$$

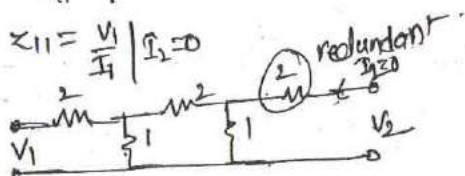
Q: (8):-



Sol:-

$$Z_{11} = ?, Z_{12} = ?$$

$$Z_{11} = \frac{V_1}{I_1} \bigg|_{I_2=0}$$



$$R_{eq} = (1||3) + 2$$

$$= \frac{3}{4} + 2 = \frac{11}{4}$$

$$\therefore \frac{V_1}{I_1} = R_{eq} = \frac{11}{4} \Omega = 2.75 \Omega$$

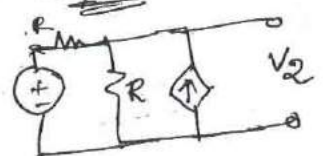
$$\begin{aligned}
 Z_{12} &= \frac{V_1}{I_2} \bigg|_{I_1=0} \\
 &= 0.95 \Omega
 \end{aligned}$$

Q: (9)

Q: (10)

Sol:-

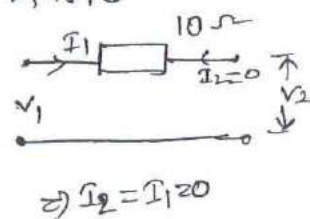
Ans: C



$$\begin{aligned}
 I_1 + \alpha I_1 &= \frac{V_2}{R} \\
 \text{where } I_1 &= \frac{V_1}{R} \\
 \Rightarrow \text{Ans: C}
 \end{aligned}$$

Q: 11  
Q: 12

Ans: d



$C=0$  ;  $A=1$

$$A = \frac{V_1}{I_2} \Big|_{I_2=0}$$

$$C = \frac{I_1}{I_2} \Big|_{I_2=0}$$

$$B = \frac{V_1}{I_2} \Big|_{V_2=0}$$

$$D = \frac{I_1}{I_2} \Big|_{V_2=0}$$

$B=10$   
 $D=1$

$$\therefore \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix}$$

Q: 13  
Q: 14

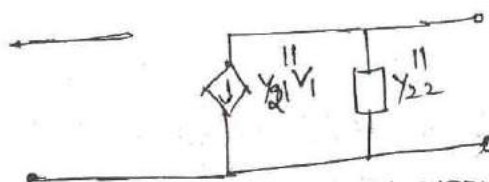
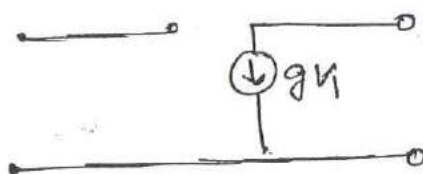
Ans: c

$$Y_{11} = Y_{22} = -Y_{12} = -Y_{21} = Y$$

$$\therefore Y_{21} = Y_{21}'' + Y_{21}'$$

$$= -Y + g$$

$Y_{21}'' = g$



Equivalent (from Y parameter)

Q: 15

$$Z(s) = \frac{\left[ 3 + \left( \frac{R_2 \times \frac{1}{j\omega C_1}}{R_2 + \frac{1}{j\omega C_1}} \right) \right] \times R_1}{\left[ 3 + \left( \frac{R_2 \times \frac{1}{j\omega C_1}}{R_2 + \frac{1}{j\omega C_1}} \right) \right] + R_1}$$

$$3 + \left[ \frac{R_2}{R_2(j\omega \frac{1}{5}) + 1} \right] R_1$$

$$\frac{3 + \left[ \frac{R_2}{R_2(j\omega \frac{1}{5}) + 1} \right] R_1}{\left[ 3 + \left( \frac{R_2}{R_2(j\omega \frac{1}{5}) + 1} \right) \right] + R_1}$$

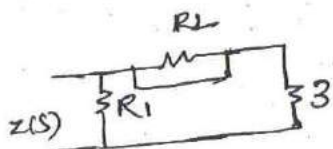
$\omega \rightarrow 0 \Rightarrow Z(s) = 2$

By substituting 'w' values

$R_1 = 6$  ;  $R_2 = 3$

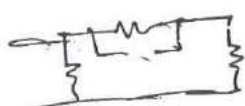
(or)

$\omega = 0$  :-



$$Z(s) = \frac{R_1 \times 3}{R_1 + 3} = 2 \Rightarrow R_1 = 6$$

$\omega = \infty$  :-



$$\Rightarrow Z = \frac{R_1 \times (3 + R_2)}{R_1 + R_2 + 3} = 3$$

$\Rightarrow R_2 = 3$

Q: 16

$$\frac{V_O(S)}{I(S)} = \frac{(100 \times 10^6 (S-1)) \times I(S)}{(10 \times 10^3 + (10 \times 10^3 \cdot S) + \frac{1}{100 \times 10^6 S}) I(S)}$$

$$= \frac{1}{(S \cdot 1 + 10^6 S^2 + 1)}$$

$$= \frac{10^6}{S^2 + 10^6 S + 1} \dots$$

Q: 17

$$Y_{11} = \frac{1}{S} + 1 = \frac{S+1}{S}$$

$$Y_{12} = -1$$

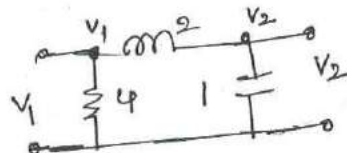
$$Y_{21} = -1$$

$$Y_{22} = S+1 \dots$$

Q: 18

sol:-

$$\frac{V_2(S)}{V_1(S)} = ?$$



$$V_2(S) = V_1(S) \frac{X_S}{\frac{1}{S} + 2S}$$

$$\frac{V_2(S)}{V_1(S)} = \frac{X_S}{\frac{1}{S} + 2S} \dots$$

$$\frac{V_1 - V_2}{2S} + \frac{V_1}{4} = 0$$

$$\frac{V_2 - V_1}{2S} + (V_2)X(S) = 0$$

$$\Rightarrow \frac{V_1 - V_2}{2S} - V_2 \cdot S = 0$$

$$\therefore \frac{V_1 - V_2}{2S} + \frac{V_1}{4} = 0$$

$$\frac{V_1 - V_2}{2S} - V_2 \cdot S = 0$$

$$\frac{V_1}{4} + V_2 S = 0$$

$$\frac{V_1}{4} = -V_2 S$$

$$\frac{V_1}{V_2} = -4S$$

$$\Rightarrow \frac{V_2}{V_1} = -\frac{1}{4S}$$

Q: 19

Ans: D

Q: 20

Ans:-  $T = T_A \cdot T_B$

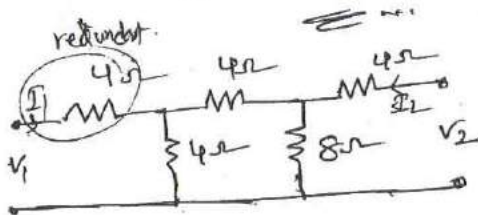
Ans:-  $\frac{C}{\dots}$

Q: 21

$$T = \begin{bmatrix} 1 & 0 \\ Y & 1 \end{bmatrix} \begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & Z \\ Y & YZ+1 \end{bmatrix}$$

Q: 22



$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

$$h_{12} = \frac{V_1}{V_2} \Big|_{I_1=0}$$

$$V_2 = I_2 R_{eq} \Rightarrow R_{eq} = 8 \Omega$$

$$V_1 = I_2 \times \frac{8}{8+8} = \left( \frac{I_2}{2} \right) \times 4$$

$$\therefore \frac{V_1}{V_2} = \frac{4 \left( \frac{I_2}{2} \right)}{8 I_2} = \frac{1}{4} = 0.25$$

Q: 23

$$I_1 = 5V_1 \times V_2$$

$$I_2 = -V_1 + V_2$$

$$Y_{11} = 5; Y_{12} = 1$$

$$Y_{21} = -1; Y_{22} = 1$$

$$Y_{12} = Y_{21} = -Y_c \Rightarrow Y_c = 1 \Omega$$

$$Y_{11} = Y_a + Y_c \Rightarrow Y_a = 5 - 1 = 4 \Omega$$

$$Y_b = 1 - 1 = 0 \dots$$

$$Y_1 = 4 \Omega$$

$$Y_3 = 1 \Omega$$

$$Y_2 = 0 \Omega \dots$$

Q: 24

Ans: c

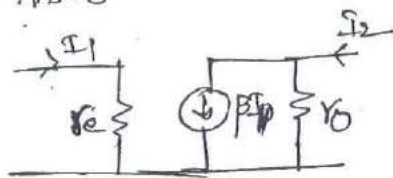
Q: 25

Ans: c → compare with standard h-parameter

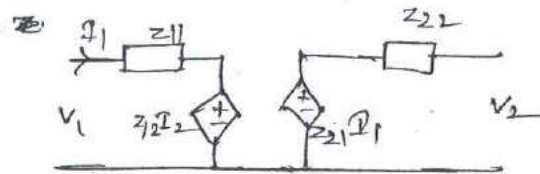
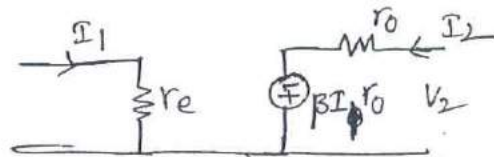
Q: 26

Ans: b

Q: 27



⇒



(OR)

$$Z_{12} = \frac{V_1}{I_2} \Big| I_1 = 0$$

$$I_1 = 0 \Rightarrow r_e I_1 = 0 \Rightarrow V_1 = 0$$

$$Z_{12} = 0$$

$$Z_{21} = \frac{V_2}{I_1} \Big| I_2 = 0$$

$$= -\frac{\beta I_1 r_0}{I_1} = -\beta r_0$$

$$Z_{12} = 0$$

$$Z_{21} = -\beta r_0$$

Q: 30

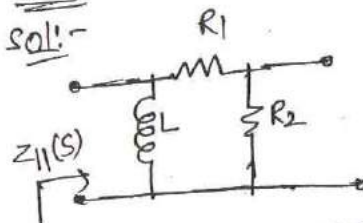
$$Z_{12} = Z_{21} = 30 \Omega$$

$$Z_{11} = 40 \Omega$$

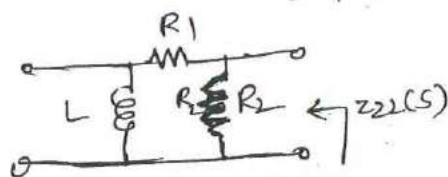
$$Z_{22} = 50 \Omega$$

Q: 31

Sol: -



$$Z_{11}(s) = \frac{Ls(R_1 + R_2)}{Ls + R_1 + R_2}$$



$$Z_{22}(s) = \frac{R_2(R_1 + Ls)}{R_1 + R_2 + Ls}$$

Ans: c

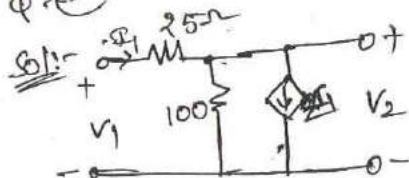
Note: - for  $Z_{11}$  &  $Z_{22}$  denominators are equal. So A & B options are eliminated.

Numerator is related to 'c', not for D, so D is wrong (just think).

Q: 29

Ans: d

Q: 32



$$V_{11} = \frac{V_1}{I_1} \Big| V_2 = 0$$

$$= 0.04 \text{ V}$$

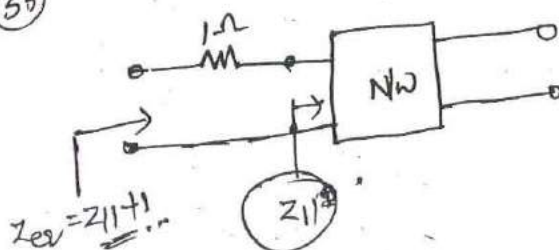
$$V_{12} = \frac{V_2}{I_1} \Big| V_1 = 0$$

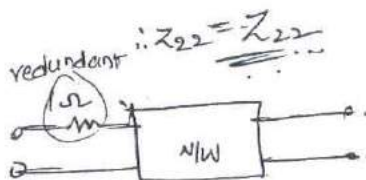
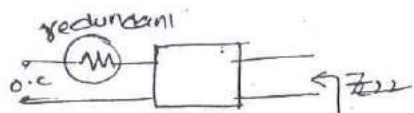
$$= -0.04 \text{ V}$$

$$V_{21} = 0.04 \text{ V}$$

$$V_{22} = 0.03 \text{ V}$$

Q: 33



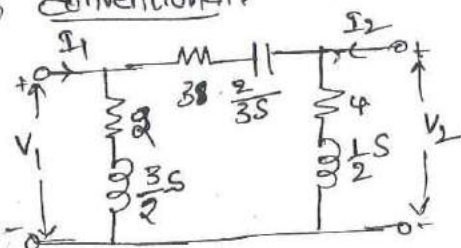


$$Z_{12} = \frac{V_1}{I_2} \mid I_1 = 0$$

since the linear element so variation of i/p & o/p are linear.

so And:  $C \dots$

conventional:-



$$Y = \begin{bmatrix} Y_a + Y_c & -Y_c \\ -Y_c & Y_b + Y_c \end{bmatrix}$$

$$Y_{12} = +Y_{21} = -Y_c$$

$$= -\frac{1}{3 + \frac{2}{3S}}$$

$$= -\frac{3S}{4S + 2} \parallel \dots$$

$$Y_{11} = Y_a + Y_c$$

$$= \frac{1}{2 + \frac{3S}{2}} + \frac{1}{3 + \frac{2}{3S}}$$

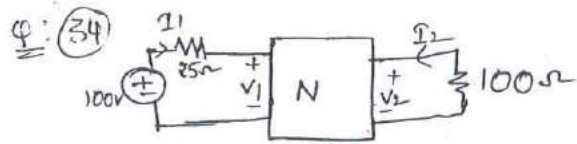
$$= \frac{2}{4 + 3S} + \frac{3S}{9S + 2}$$

$$= \frac{18S + 4 + 12S + 9S^2}{(4 + 3S)(9S + 2)} \parallel \dots$$

$$Y_{22} = Y_b + Y_c$$

$$= \frac{1}{4 + \frac{S}{2}} + \frac{1}{3 + \frac{2}{3S}}$$

$$= \frac{2(1)}{8 + S} + \frac{3S}{9S + 2} \parallel \dots$$



$$V_2 = -100 I_2$$

$$\Rightarrow I_2 = -\frac{V_2}{100}$$

Y-param eqns:

$$I_1 = Y_{11} V_1 + Y_{12} V_2$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2$$

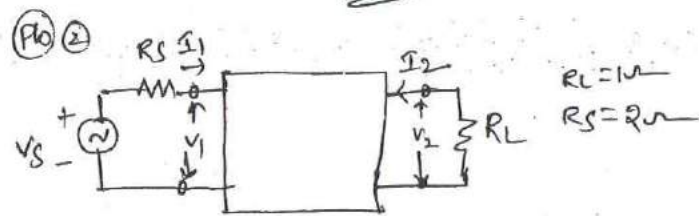
$$I_2 = 0.01 V_1 + 0.1 V_2$$

$$-\frac{V_2}{100} = 0.01 V_1 + 0.1 V_2$$

$$\Rightarrow 0.01 V_1 + V_2 [0.1 + 0.01] = 0$$

$$\dots \dots \dots$$

$$\frac{V_2}{V_1} = -\frac{1}{11}$$



$$V_S = V_1 + I_1 R_S$$

$$V_1 = V_S - I_1 R_S$$

$$V_1(s) = \frac{1}{s} - (I_1(s) \cdot 2) \dots (1)$$

$$V_2 = -I_2 R_L$$

$$V_2(s) = -I_2(s) \times 1 \dots (2)$$

Given data

$$[Z] = \begin{bmatrix} \frac{2}{s+1} & \frac{1}{s+1} \\ \frac{1}{s+1} & \frac{6}{s+1} \end{bmatrix}$$

$$\begin{bmatrix} V_1(s) \\ V_2(s) \end{bmatrix} = [Z] \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix}$$

$$V_1(s) = \frac{2}{s+1} I_1(s) + \frac{1}{s+1} I_2(s)$$

$$V_2(s) = \frac{1}{s+1} I_1(s) + \frac{6}{s+1} I_2(s)$$

but

$$V_1(s) = \frac{1}{s} - I_1(s) \cdot 2$$

$$V_2(s) = -I_2(s) \dots$$

so

$$\frac{1}{s} - 2I_1(s) = \frac{2}{s+1} I_1(s) + \frac{1}{s+1} I_2(s)$$

$$-I_2(s) = \frac{1}{s+1} I_1(s) + \frac{6}{s+1} I_2(s)$$

$$\Rightarrow I_1(s) = \dots, I_2(s) = \dots$$

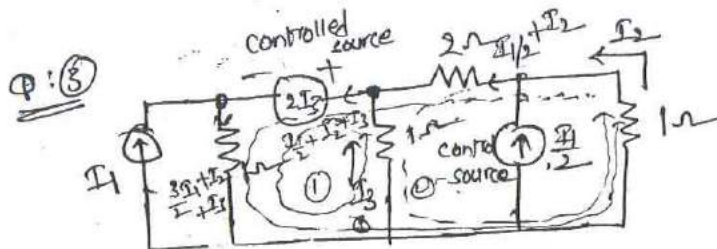
$$I_2(s) = \underline{\hspace{2cm}}$$

$$V_2(s) = -1 \times I_2(s) \quad (08)$$

$$V_2(t) = -\mathcal{L}^{-1}(I_2(s))$$

$$= 0.037 + 0.45 e^{-1.9t} - 0.083 e^{-7.08t}$$

Note:-  
zero state means without  
initial values.



$$\frac{I_2}{I_1} = ?$$

Based on KCL we concluded all the  
branch currents.

Now go for KVL:- (Mechanical is)

for loop (1):

$$(2I_3) + 1 \left( \frac{3I_1 + I_2}{2} + I_3 \right) + I_3 = 0 \rightarrow (1)$$

for loop (2):

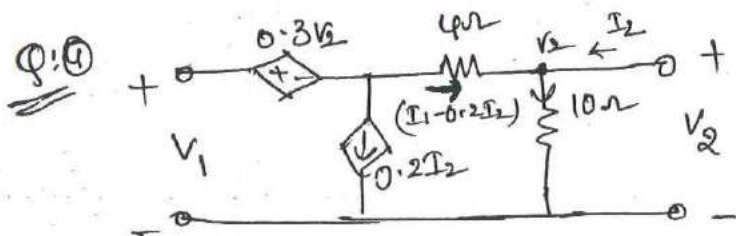
$$2 \left( \frac{I_1}{2} + I_2 \right) + 1(I_3) + I_2 = 0 \rightarrow (2)$$

for loop (3):

$$2 \left( \frac{I_1}{2} + I_2 \right) + 2I_3 + 1(3I_1 + I_2 + I_3) + I_2(1) = 0 \rightarrow (3)$$

from (1) & (2) & (3)

$$\frac{I_2}{I_1} = -\frac{11}{26} \text{ Ans.}$$



$$V_1 = h_{11} I_1 + V_2 \cdot h_{12}$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

$$h_{12} = \frac{V_1}{V_2} \Big|_{I_1=0}$$

$$h_{22} = \frac{I_2}{V_2} \Big|_{I_1=0}$$

$$\left. \begin{aligned} V_1 &= \frac{2}{s+1} I_1(s) + \frac{1}{s+1} I_2(s) \\ V_2 &= \frac{1}{s+1} I_1(s) + \frac{6}{s+1} I_2(s) \end{aligned} \right\} \begin{aligned} &\rightarrow (1) \\ &\rightarrow (2) \end{aligned}$$

but  $V_2(s) = -I_2(s) R_L = -I_2(s) \times 1$

$$V_1(s) = V_s - I_1(s) R_s \rightarrow (3)$$

$$V_1(s) = \frac{1}{s} - I_1(s) \times 2 \rightarrow (4)$$

from (1) & (2) & (3) & (4)

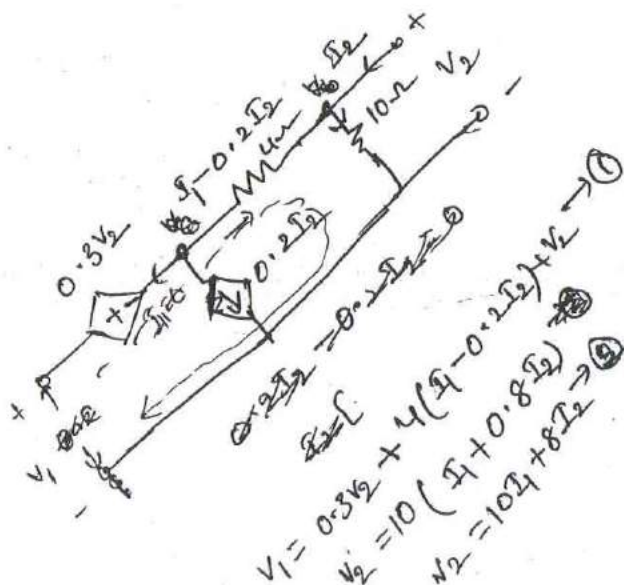
$$I_1(s) = \underline{\hspace{2cm}}$$

$$I_2(s) = \underline{\hspace{2cm}}$$

$$\therefore V_2(s) = -I_2(s) \times 1$$

$$V_2(t) = -\mathcal{L}^{-1}(I_2(s))$$

Ans:  $-\frac{11}{26} \frac{I_2}{I_1}$



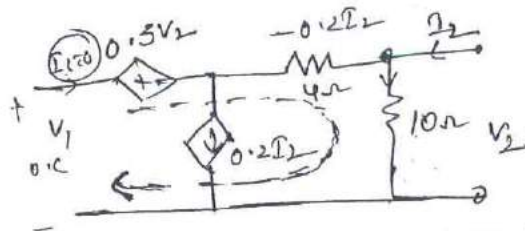
from (1) & (2)

$$V_1 = I_1 + I_2$$

$$V_2 = I_1 + I_2$$

Ans:  $Z_{12} = 9.6$

$h_{12} = \frac{V_1}{V_2} \bigg|_{I_1=0}$



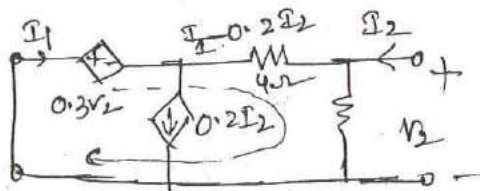
$$0.3V_2 - V_1 - 4(0.2I_2) + 10(I_2 - 0.2I_2) = 0 \quad \text{--- (1)}$$

But  $V_2 = 10(I_2 - 0.2I_2)$  --- (2)

from (1) & (2)

$h_{12} = \dots$

$Y_{12} = \frac{I_1}{V_2} \bigg|_{V_1=0}$

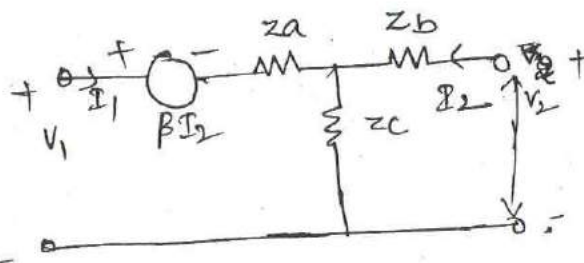


$$+0.3V_2 + 4(I_1 - 0.2I_2) + 10(0.8I_2) = 0 \quad \text{--- (1)}$$

$V_2 = 10(I_1 + 0.8I_2)$  --- (2)

from (1) & (2)  $Y_{12} = \dots$

Q.5



By mesh analysis

we can find  $Z_a, Z_b, Z_c$

Since  $Z_a = Z_{11} - Z_{12}$

$Z_c = Z_{12} = Z_{21}$

$Z_b = Z_{22} - Z_{12}$

filter is  
workbook

$$T.F = \frac{1}{(1+sT)} \cdot \frac{1}{(1+sT)}$$

roots are real & equal, so critically damped system,

Q:1

Q:2

Ans: a

Q:3

Ans: d

Q:4

Ans: d

Q:5

Ans: c

Q:6

Ans: b

Q:7

Ans: b

Q:8

Ans: c

Q:9

$$B_1 = \frac{R}{L} = \omega_2 - \omega_1$$

$$B_2 = \frac{R}{L} = \frac{4R}{L}$$

$$\Rightarrow \frac{B_1}{B_2} = \frac{1}{4}$$

Q:10

Ans: d

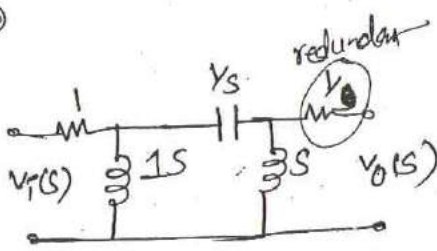
### Network Synthesis

Q:1

RC integrator  $\rightarrow$  low pass  $= \frac{1}{RCs+1}$

$$\frac{1}{RC(s + \frac{1}{RC})}$$

Q:2



finite pole & zero at  $\infty$ .  
NOTE: total no. of poles = total no. of zeros  
Ans: d

zero at  $\infty$ .  
of poles = total no. of zeros

$$\Rightarrow Z_{eq} = \frac{(s+1) \times s}{s} + 1 = \frac{s(s^2+1)}{s^2+1+s^2} + 1 = \frac{2s^3+2s^2+s+1}{2s^2+1}$$

OP = open

$$\text{Ans: } Z_{11} = \frac{s^3+2s^2+s+1}{2s^2+1} = Z_{22}$$

Q:3

$$Y(s) = \frac{s^2+2.5s+1}{s^2+4s+3} = \frac{(s+0.5)(s+2)}{(s+1)(s+3)}$$

properties of  $Y_{RL}$  &  $Z_{RC}$  are same  $\rightarrow$  lowest freq is  $\rightarrow$  pole is wrong.  
so first statement is wrong.

2nd statement

3 x  
4 ✓

$$Y(0) = \frac{0.5 \times 2}{1 \times 3} = \frac{1}{3}$$

Ans: b

Q: ②

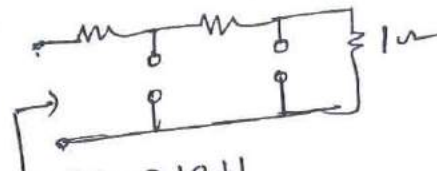
$$Z(s) = \left( \left( \frac{\frac{9}{2s} \parallel 1}{\frac{9}{2s} + 1} \right) + R \right) \parallel \left( \frac{1}{Cs} \right) + 3$$

Ans: - ②

Given that

$$Z(s) = \frac{3(s^2 + 6s + 8)}{Cs^2 + 4s + 3}$$

$$s=0 \rightarrow Z(s) = 8 = Z(0)$$



$$Z(0) = 3 + R + 1$$

$$\Rightarrow 3 + R + 1 = 8 \Rightarrow R = 4$$

Q: ③

$$Z(s) = \frac{1}{s} + \frac{1}{s + \frac{1}{\tau}}$$

$$I(s) = \frac{1}{s}$$

$$V(s) = \frac{I(s)}{Y(s)}$$

$$= \frac{\frac{1}{s}}{\frac{1}{s} + \frac{1}{s + \frac{1}{\tau}}}$$

$$Z(s) = \frac{V(s)}{I(s)} = \frac{\frac{1}{s}}{\frac{1}{s} + \frac{1}{s + \frac{1}{\tau}}} = \frac{2s + \frac{1}{\tau}}{s + \frac{1}{\tau}}$$

$$Y(s) = \frac{s + \frac{1}{\tau}}{2s + \frac{1}{\tau}}$$

$$= \frac{1}{2} + \frac{\frac{1}{2\tau}}{2s + 1}$$

$$\downarrow \quad \downarrow$$

$$R = 2\Omega \quad Z_a = \frac{1}{2\tau} \rightarrow L = 2\tau$$

$$Z_b = 1$$

$$\downarrow$$

$$R$$

Ans: (d)

Q: ⑥:-

$$Z(s) = s + 2 + \frac{3}{s}$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$L=1 \quad R=2 \quad C=\frac{1}{3}$$

$$\text{If } Z(s) = \text{PRF} \Rightarrow Y(s) = \text{PRF}$$

Ans: ⑥

NOTE:- If  $Z(s)$  is positive real fun.  
 $\frac{1}{Z(s)}$  also positive real fun.